Algebra Mastery Pack — Practice Test A

Grade 11 algebra, functions & data analysis · 24 questions · suggested time 45 minutes · no calculator needed

Name: _____ Date: _____
Score: _____ / 24

Instructions. Write full working in the space under each question; answers alone do not show how you got there. Circle one letter for a multiple-choice question and write a single value on the answer line for a short-answer question. Answer every question — a blank scores zero. The answer key with a worked solution for every question is a separate PDF; mark your own paper only after you finish.

1. [Solving Linear Equations (for x)]

Solve $2x + 6 = 14$.

A) $x = 3$
B) $x = 4$
C) $x = 8$
D) $x = 10$

2. [Literal Equations & Solving for y]

Solve $y = mx + b$ for b.

A) $b = y - mx$
B) $b = y + mx$
C) $b = mx - y$
D) $b = y \div mx$

3. [Linear Inequalities]

Solve $x + 3 < 7$.

A) $x < 4$
B) $x < 10$
C) $x > 4$
D) $x \leq 4$

4. [Systems of Linear Equations]

Solve the system: $y = 2x$ and $x + y = 9$.

A) (3, 6)
B) (6, 3)
C) (3, 3)
D) (4, 5)

5. [Functions: Notation, Domain & Range]

If $f(x) = 3x + 1$, what is $f(2)$?

A) 5
B) 6
C) 7
D) 8

---

6. [Linear Functions: Slope, Intercepts & Equations]

In the equation $y = 2x + 1$, what is the slope?

A) 1
B) 2
C) $2x$
D) 0

---

7. [Quadratic Functions & Equations]

Solve $x^2 + 5x + 6 = 0$ by factoring.

A) $x = -2$ or -3
B) $x = 2$ or 3
C) $x = -2$ or 3
D) $x = 5$ or 6

---

8. [Polynomials: Operations & Factoring]

Add: $(3x^2 + 2x + 1) + (x^2 - x + 4)$.

A) $4x^2 + x + 5$
B) $4x^2 + 3x + 5$
C) $3x^2 + x + 5$
D) $4x^2 + x + 3$

---

9. [Radical & Rational Expressions]

Simplify $\sqrt{50}$.

A) $5\sqrt{2}$
B) 25
C) $10\sqrt{5}$
D) $2\sqrt{25}$

---

10. [Exponential Functions & Growth/Decay]

In $y = 3 \cdot 2^x$, what is the initial value (the value when $x = 0$)?

A) 3
B) 2
C) 6
D) 0

---

11. [Data: Measures of Center & Spread]

Find the mean of {2, 4, 4, 6, 9}. Enter a number.

Answer: _____

12. [Scatter Plots, Line of Best Fit & Correlation]

If y tends to rise as x rises, the correlation is ___.

A) positive
B) negative
C) none
D) undefined

---

13. [Absolute Value Equations & Inequalities]

Solve $|x| = 7$.

A) $x = 7$ only
B) $x = -7$ only
C) $x = 7$ or $x = -7$
D) No solution

---

14. [Arithmetic & Geometric Sequences]

What is the next term of the arithmetic sequence 4, 7, 10, 13, …?

A) 14
B) 15
C) 16
D) 17

---

15. [Inverse Functions]

If $f(2) = 9$, what is $f^{-1}(9)$?

A) 2
B) 9
C) 1/9
D) -9

---

16. [Function Transformations & Vertex Form]

The graph of $y = f(x) + 3$ is the graph of $y = f(x)$…

A) shifted up 3 units
B) shifted down 3 units
C) shifted right 3 units
D) stretched vertically by 3

---

17. [Systems of Linear Inequalities & Linear Programming]

Which point is NOT a solution of the system $y > x - 1$ and $y \leq 2x$?

A) (1, 2)
B) (0, 0)
C) (2, 3)
D) (−1, 0)

---

18. [Probability from Two-Way Tables]

Survey of 200 students: how they travel and whether they arrived on time

	On time	Late	Total
Rides the bus	60	20	80
Walks	90	30	120
Total	150	50	200

Using the table, what is P(a student rides the bus)?

A) 0.30
B) 0.40
C) 0.75
D) 0.80

---

19. [Quadratic Functions & Equations]

Solve $x^2 - 4x - 5 = 0$ by factoring.

A) $x = 5$ or -1
B) $x = -5$ or 1
C) $x = 5$ or 1
D) $x = -4$ or 5

---

20. [Polynomials: Operations & Factoring]

Multiply: $(2x)(3x)$.

A) $6x^2$
B) $5x^2$
C) $6x$
D) $5x$

---

21. [Radical & Rational Expressions]

Solve $\sqrt{x} = 5$. Enter the value of x.

Answer: _____

22. [Exponential Functions & Growth/Decay]

For $y = 2^x$, find y when $x = 3$. Enter a number.

Answer: _____

23. [Data: Measures of Center & Spread]

Find the median of {2, 4, 4, 6, 9}. Enter a number.

Answer: _____

24. [Scatter Plots, Line of Best Fit & Correlation]

If y tends to fall as x rises, the correlation is ___.

A) negative
B) positive
C) none
D) undefined
